Unit III: Cell biology and Biomolecules

Chapter 6

Cell: The Unit of Life

Learning Objectives

The learner will be able to,

- *Understand the ideas of cell theory and the different concepts associated with it*
- *Recognize the basic structure of cell and differentiate the cells of animals, plants, bacteria and viruses*
- *Explain the structure and functions of cell organelles including nucleus*
- *Recognize the structure of chromosome and its types*

Chapter Outline

6.1. Discovery
6.2. Microscopy
6.3. Cell theory
6.4. Types of Cell
6.5. Plant and Animal cell
6.6. Cell organelles
6.7. Nucleus
6.8. Flagella

The word '*cell*' comes from the Latin word '*Celle*" which means 'a small compartment'. The word cell was first used by Robert Hooke (1662) therefore the term '*cell*' is as old as 300 years.

6.1. Discovery

Aristotle (384-322BC), was the one who first recognised that animals and plants consists of organised structural units but unable to explain what it was. In 1660's **Robert Hooke** observed something which looks like 'honeycomb with a great numbers of little boxes' which was later called as '**cell**' from the cork tissue. In 1665, He compiled his work as **Micrographia**. Later, **Anton Van Leeuwenhoek** observed unicellular particles which he named as '***animalcules***'. **Robert Brown** (1831-39) described the spherical body in plant cell as nucleus. **H. J. Dutrochet** (1824), a French scientist, was the first to give an idea on cell theory. Later, **Matthias Schleiden** (German Botanist) and **Theodor Schwann** (German Zoologist) (1833) outlined the basic features of the cell theory. **Rudolf Virchow** (1858) explained the cell theory by adding a feature stating that all living cells arise from pre-existing living cells by 'cell division'.

Scientist



Aristotle (384–322BC) | Robert Hooke (1635–1703) | Antonie von Leeuwenhoek (1632–1723)

Schleiden (1804–1881) & Schwann (1810–1882) | Rudolf Virchow (1821–1902)

Figure 6.1

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